

Noel Sason

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EDUCATION

University of California, Berkeley — *B.A. Molecular Biology | B.S. Data Science*

Aug 2024 – May 2028

GPA: 3.8 / 4.0 | **Relevant Coursework:** CS61B (Data Structures & Algorithms), Math 54 (Linear Algebra), Data 8 (Foundations of Data Science — Statistics & Probability), Data C88C (Computational Structures in Data Science)

EXPERIENCE

Research Affiliate — *Lawrence Berkeley National Laboratory*

Jan 2026 – Present

- Enabled cross-cohort meta-analysis across 80+ disease–microbiome studies by building a DuckDB-backed pipeline over the KG-Microbe knowledge graph for SQL-based traversal of taxa, chemicals (CHEBI), environments (ENVO), and phenotypic traits.
- Accelerated study onboarding for 4 disease cohorts (colorectal cancer, lung cancer, epilepsy, skin wounds) by writing scripts that transform BugSigDB differential-abundance exports into per-study directories of taxa lists keyed to NCBI Taxon IDs.
- Reduced manual data-wrangling time by designing a YAML-driven trait-profile generator that parses results across all study directories and emits wide and long TSV tables (study × phenotype) for downstream statistical analysis.
- Improved team reproducibility, as measured by zero environment-setup issues across contributors, by standardizing workflows with Justfile task runners, uv-managed Python environments, and PR-based collaboration on GitHub.

Finance Director — *Various Roles within the ASUC*

Sept 2024 – Present

- Administered approximately \$200,000 in student organization funding, as measured by on-time grant disbursements and zero compliance violations, by managing CASSA allocations and awarding grants to student-led proposals across two orgs.

PROJECTS

Canvascope, Lectra & PersonalGraph MCP — canvascope.org

Founder / CEO / Lead Developer

- Grew Canvascope to 46 active users across 10+ university Canvas domains by founding a privacy-first AI study assistant (Chrome extension, Manifest V3) with GCP OAuth authentication, local-first indexing, and sub-100ms semantic search.
- Achieved citation-grounded AI responses, as measured by every answer linking to its source document, by designing a RAG pipeline that indexes course PDFs, pages, and assignments into vector embeddings and retrieves relevant context at query time.
- Reduced cloud inference costs and improved user privacy by integrating Chrome’s built-in Gemini Nano for on-device AI and building personalized learning workflows that adapt to a student’s academic background and learning style.
- Enabled cross-device study workflows by developing Lectra (iOS/iPad) with Apple Pencil annotation, DropBridge — a custom high-speed cross-device file-transfer protocol — and a Supabase backend for real-time sync of annotated PDFs.
- Improved search relevance and UI engagement, as measured by A/B experiments across 10+ universities, by conducting user interviews and usability studies and iterating on ranking, categorization, and interface design.
- Built PersonalGraph, a TypeScript MCP server that gives AI assistants a persistent personal knowledge graph — 15 tools/resources, a four-tier visibility lattice with secret redaction and sensitive-topic auto-classification, crash-safe atomic writes, and an append-only audit log.

RxBrief — rxbrief.org

Co-Founder / CTO

- Delivered instant drug-label access and consolidated job discovery from 3 sources by building a full-stack platform with a live FDA API, real-time Supabase chat, and client–server REST integrations (Greenhouse, Lever, RSS) with server-side filtering.
- Translated clinical workflows into engineering requirements by interviewing healthcare professionals, directly shaping product scope and task prioritization.

SIS Community App — *South Indian Society, UC Berkeley*

Co-President

- Onboarded 40 beta testers from a 200+ member student organization by developing a native iOS app with Stripe-integrated event management, a Cloudflare R2–backed shared photo album, and real-time messaging for member engagement.
- Replaced 4+ fragmented tools with a single platform, as measured by leadership adoption for all org operations, by architecting a Supabase backend with role-based access control, committee-scoped file archives, and push notifications.

TECHNICAL SKILLS & CERTIFICATIONS

Languages: Python, Java, Swift/SwiftUI, JavaScript, SQL, HTML5/CSS

ML & Data: PyTorch, scikit-learn, Pandas, NumPy, DuckDB, MongoDB, embeddings & vector search

Quant & Methods: Linear algebra, statistics & probability, A/B testing, data structures & algorithms

Web & Mobile: Node.js, Next.js, React, iOS/SwiftUI, Xcode, REST & client–server API design

Tools & Platforms: Git, GitHub Actions, Docker, Linux/Unix, Supabase, Vercel, Stripe, Cloudflare R2 **Cert:** AHA BLS